There exists $\eta > 0$ such that $\boldsymbol{\theta}_i \xrightarrow{t\to\infty} \tilde{\boldsymbol{\theta}}_i$ for all i, where

$$\begin{aligned}
\tilde{\boldsymbol{\theta}}_1 &= \arg\min_{\boldsymbol{\theta}_1} \ell_1(\boldsymbol{\theta}_1) \\
\tilde{\boldsymbol{\theta}}_2 &= \arg\min_{\boldsymbol{\theta}_2} \ell_2(\tilde{\boldsymbol{\theta}}_1, \boldsymbol{\theta}_2) \\
&\dots \\
\tilde{\boldsymbol{\theta}}_n &= \arg\min_{\boldsymbol{\theta}_n} \ell_n(\tilde{\boldsymbol{\theta}}_1, \tilde{\boldsymbol{\theta}}_2, \dots, \tilde{\boldsymbol{\theta}}_{n-1}, \boldsymbol{\theta}_n).
\end{aligned}$$

Proof. Let $\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_n$ be as in the theorem statement. We proceed by induction. For the base case, note that the myopic descent iterates for $\boldsymbol{\theta}_1^{(t)}$ are independent of $\boldsymbol{\theta}_j^{(t)}$ for $j > i$. Thus, the standard convergence theorem gives that $\boldsymbol{\theta}_1^{(t)} \to \tilde{\boldsymbol{\theta}}_1$ as $t \to \infty$.

Now, assume $\boldsymbol{\theta}_i^{(t)} \xrightarrow{t\to\infty} \tilde{\boldsymbol{\theta}}_i$ for all $i < k$. Thus, for any $\varepsilon > 0$, for sufficiently large t,

$$\|(\boldsymbol{\theta}_i^{(t)})_{i=1}^{k-1} - (\tilde{\boldsymbol{\theta}}_i)_{i=1}^{k-1}\|_2 < \varepsilon.$$

Hence, since ℓ_k is L-smooth, for any $\boldsymbol{\theta}_k$,

$$\|\nabla_{\boldsymbol{\theta}_k} \ell_k(\boldsymbol{\theta}_1^{(t)}, \dots, \boldsymbol{\theta}_{k-1}^{(t)}, \boldsymbol{\theta}_k) - \nabla_{\boldsymbol{\theta}_k} \ell_k(\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_{k-1}, \boldsymbol{\theta}_k)\|_2 < L\varepsilon.$$

Expanding and rearranging,

$$\langle \nabla_{\boldsymbol{\theta}_k} \ell_k(\boldsymbol{\theta}_1^{(t)}, \dots, \boldsymbol{\theta}_{k-1}^{(t)}, \boldsymbol{\theta}_k), \nabla_{\boldsymbol{\theta}_k} \ell_k(\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_{k-1}, \boldsymbol{\theta}_k) \rangle \geq \frac{1}{2}\|\nabla_{\boldsymbol{\theta}_k} \ell_k(\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_{k-1}, \boldsymbol{\theta}_k)\|_2^2 - \frac{1}{2}L^2\varepsilon^2$$

is bounded away from zero as long as, say,

$$\|\nabla_{\boldsymbol{\theta}_k} \ell_k(\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_{k-1}, \boldsymbol{\theta}_k)\|_2 \geq \sigma\|\boldsymbol{\theta}_k - \tilde{\boldsymbol{\theta}}_k\|_2 > (L+1)\varepsilon,$$

using the σ-strong convexity of ℓ_k. That is, as long as $\|\boldsymbol{\theta}_k - \tilde{\boldsymbol{\theta}}_k\|_2 > \dfrac{(L+1)\varepsilon}{\sigma}$, it is guaranteed that $-\nabla_{\boldsymbol{\theta}_k} \ell_k(\boldsymbol{\theta}_1^{(t)}, \dots, \boldsymbol{\theta}_{k-1}^{(t)}, \boldsymbol{\theta}_k)$ is a descent direction for $\ell_k(\tilde{\boldsymbol{\theta}}_1, \dots, \tilde{\boldsymbol{\theta}}_{k-1}, \boldsymbol{\theta}_k)$. This is a sufficient condition for $\boldsymbol{\theta}_k$ to converge to a $\dfrac{(L+1)\varepsilon}{\sigma}$-neighborhood of $\tilde{\boldsymbol{\theta}}_k$ given a small enough step size (Beck, 2017). Since $\varepsilon > 0$ is arbitrary, this completes the inductive step. □

C.4 Myopic descent with tied weights

Definition 11. *A function $f(\boldsymbol{x}, \boldsymbol{y})\colon \mathbb{R}^a \times \mathbb{R}^a \to \mathbb{R}$ with continuous second derivatives is ρ-**forward-biased** if, for all $\boldsymbol{y} \in \mathbb{R}^a$ and $\boldsymbol{x} = \boldsymbol{y}$,*

$$\boldsymbol{H}_{\boldsymbol{y},\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y}) + \boldsymbol{H}_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y}) \succ 0$$

and

$$\kappa(\boldsymbol{H}_{\boldsymbol{y},\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y}) + \boldsymbol{H}_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y})) < \rho,$$

where κ is the condition number, and we write the Hessian of f as a block matrix:

$$\boldsymbol{H} f(\boldsymbol{x}, \boldsymbol{y}) = \begin{bmatrix} \boldsymbol{H}_{\boldsymbol{x},\boldsymbol{x}} f(\boldsymbol{x}, \boldsymbol{y}) & \boldsymbol{H}_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y}) \\ \boldsymbol{H}_{\boldsymbol{y},\boldsymbol{x}} f(\boldsymbol{x}, \boldsymbol{y}) & \boldsymbol{H}_{\boldsymbol{y},\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y}) \end{bmatrix} = \begin{bmatrix} \left(\dfrac{\partial f(\boldsymbol{x}, \boldsymbol{y})}{\partial x_i \partial x_j}\right)_{\substack{1\le i\le a \\ 1\le j\le a}} & \left(\dfrac{\partial f(\boldsymbol{x}, \boldsymbol{y})}{\partial x_i \partial y_j}\right)_{\substack{1\le i\le a \\ 1\le j\le b}} \\ \left(\dfrac{\partial f(\boldsymbol{x}, \boldsymbol{y})}{\partial y_i \partial x_j}\right)_{\substack{1\le i\le b \\ 1\le j\le a}} & \left(\dfrac{\partial f(\boldsymbol{x}, \boldsymbol{y})}{\partial y_i \partial y_j}\right)_{\substack{1\le i\le b \\ 1\le j\le b}} \end{bmatrix}.$$

Lemma 12. *If $\boldsymbol{A} \in \mathbb{R}^{a\times a}$ is positive definite with $\kappa(A) < \rho$, then there exists some $\eta > 0$ such that $\boldsymbol{I} - \eta\boldsymbol{A}$ is a $(1 - \rho^{-1})$-contraction. Explicitly, for any $\boldsymbol{y} \in \mathbb{R}^a$,*

$$\|(\boldsymbol{I} - \eta\boldsymbol{A})\boldsymbol{y}\|_2 \leq (1 - \rho^{-1})\|\boldsymbol{y}\|_2.$$

Proof. Set $\eta = 1/\lambda_{\max}(\boldsymbol{A})$. Then $\boldsymbol{I} - \eta\boldsymbol{A} \succeq 0$ with

$$\lambda_{\max}(\boldsymbol{I} - \eta\boldsymbol{A}) = 1 - \eta\lambda_{\min}(\boldsymbol{A}) = 1 - \frac{\lambda_{\min}(\boldsymbol{A})}{\lambda_{\max}(\boldsymbol{A})} < 1 - \rho^{-1}.$$

It immediately follows that $\boldsymbol{I} - \eta\boldsymbol{A}$ is a $(1-\rho^{-1})$-contraction. □

Theorem 13. *Let $f(\boldsymbol{x}, \boldsymbol{y})\colon \mathbb{R}^a \times \mathbb{R}^a \to \mathbb{R}$ be ρ-forward biased, σ-strongly convex, and L-smooth with continuous second derivatives for some $\rho, \sigma, L > 0$. Then, there exists $\tilde{\boldsymbol{y}} \in \mathbb{R}^a$ such that*

$$\tilde{\boldsymbol{y}} = \arg\min_{\boldsymbol{y} \in \mathbb{R}^a} f(\tilde{\boldsymbol{y}}, \boldsymbol{y}). \tag{4}$$

Further, for some step size $\eta > 0$, the iterates of myopic gradient descent with tied weights

$$\boldsymbol{y}^{(t+1)} = \boldsymbol{y}^{(t)} - \eta \nabla_{\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y})|_{\boldsymbol{x}=\boldsymbol{y}=\boldsymbol{y}^{(t)}}$$

converge to $\tilde{\boldsymbol{y}} \in \mathbb{R}^a$ satisfying (4). We call such $\tilde{\boldsymbol{y}}$ a myopic solution.

We then recover the original sequential modeling setting by defining $f(\tilde{\boldsymbol{\theta}}, \boldsymbol{\theta}) := \sum_{i=1}^{n} \ell_i(\tilde{\boldsymbol{\theta}}, \ldots, \tilde{\boldsymbol{\theta}}, \boldsymbol{\theta})$.

Proof. First, note that if myopic descent does converge, it must be to a point $\tilde{\boldsymbol{y}}$ such that $\tilde{\boldsymbol{y}} = \arg\min_{\boldsymbol{y}} f(\tilde{\boldsymbol{y}}, \boldsymbol{y})$. Indeed, at convergence we must have $\nabla_{\boldsymbol{y}} f(\tilde{\boldsymbol{y}}, \boldsymbol{y})|_{\boldsymbol{y}=\tilde{\boldsymbol{y}}} = 0$, so strong convexity tells us that $\tilde{\boldsymbol{y}}$ is optimal. Thus, if we establish that myopic descent with small enough step size $\eta > 0$ converges to some $\tilde{\boldsymbol{y}}$, we automatically get the existence of a myopic solution. For this, it suffices to show that the gradient descent step

$$g_\eta(\boldsymbol{y}) = \boldsymbol{y} - \eta \nabla_{\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y})|_{\boldsymbol{x}=\boldsymbol{y}}$$

is strictly contractive, so that iterates of g_η converge to a fixed point.

Consider arbitrary $\boldsymbol{y}$ and $\boldsymbol{y}'$. Then, by chain rule and the mean value theorem applied to the map $\boldsymbol{y}' \mapsto \nabla_{\boldsymbol{y}} f(\boldsymbol{x}, \boldsymbol{y})|_{\boldsymbol{x}=\boldsymbol{y}=\boldsymbol{y}'}$,

$$\nabla_{\boldsymbol{y}} f(\boldsymbol{y}', \boldsymbol{y}') - \nabla_{\boldsymbol{y}} f(\boldsymbol{y}, \boldsymbol{y}) = (\boldsymbol{H}_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{y}'', \boldsymbol{y}'') + \boldsymbol{H}_{\boldsymbol{y},\boldsymbol{y}} f(\boldsymbol{y}'', \boldsymbol{y}''))(\boldsymbol{y}' - \boldsymbol{y}) \tag{5}$$

for some $\boldsymbol{y}'' \in [\boldsymbol{y}, \boldsymbol{y}']$. Using the definition of g_η and substituting in (5),

$$\begin{aligned}
\|g_\eta(\boldsymbol{y}') - g_\eta(\boldsymbol{y})\|_2^2 &= \|(\boldsymbol{y}' - \boldsymbol{y}) - \eta(\nabla_{\boldsymbol{y}} f(\boldsymbol{y}', \boldsymbol{y}') - \nabla_{\boldsymbol{y}} f(\boldsymbol{y}, \boldsymbol{y}))\|_2^2 \\
&= \|(I - (\boldsymbol{H}_{\boldsymbol{x},\boldsymbol{y}} f(\boldsymbol{y}'', \boldsymbol{y}'') + \boldsymbol{H}_{\boldsymbol{y},\boldsymbol{y}} f(\boldsymbol{y}'', \boldsymbol{y}'')))(\boldsymbol{y}' - \boldsymbol{y})\|_2^2 \\
&\leq (1 - \rho^{-1})(\boldsymbol{y}' - \boldsymbol{y}),
\end{aligned}$$

where the last line is by ρ-forward bias and Lemma 12. That is, g_η is $(1-\rho^{-1})$-contractive, completing the proof. □

C.5 Properties of sine

Lemma 14. *Let* $\mathrm{x} \sim \mathcal{N}(0, 1)$. *Then*

$$\mathrm{Var}(\sin(b\mathrm{x})) = \frac{1}{2} - \frac{e^{2b^2}}{2}.$$

Proof.

$$\mathrm{Var}(\sin(b\mathrm{x})) = (2\pi)^{-1/2} \int_{-\infty}^{\infty} \sin^2(bx) e^{-x^2/2}\, dx = \frac{1}{2} - \frac{e^{2b^2}}{2}.$$

□

Lemma 15. *Let* $\mathrm{x} \sim \mathcal{N}(0, 1)$. *Then*

$$\rho(\mathrm{x}, \sin(b\mathrm{x})) = \frac{2be^{3b^2/2}}{e^{2b^2} - 1} \in O(e^{-b^2/2}),$$

where ρ is the Pearson correlation coefficient.

Proof. By symmetry, $\mathbb{E}[\sin(b\mathrm{x})] = 0$. We calculate

$$\mathrm{Cov}(\mathrm{x}, \sin(b\mathrm{x})) = (2\pi)^{-1/2} \int_{-\infty}^{\infty} x \sin(bx) e^{-x^2/2}\, dx = be^{-b^2/2}.$$

We already computed the variance in Lemma 14, so

$$\rho(\mathrm{x}, \sin(b\mathrm{x})) = \frac{\mathrm{Cov}(\mathrm{x}, \sin(b\mathrm{x}))}{\sqrt{\mathrm{Var}(\mathrm{x})\mathrm{Var}(\sin(b\mathrm{x}))}} = \frac{2be^{3b^2/2}}{e^{2b^2} - 1}.$$

In our experiments we set $b = 10$, so $\rho(\mathrm{x}, \sin(b\mathrm{x})) < 10^{-20}$.

D Details and additional experiments

D.1 Synthetic setting: $\mathcal{D}_p$ task

We use a smaller version of the GPT-2 architecture, adapted to regression tasks. That is, the token embedding and unembedding layers are replaced with a trained linear map from the input space to the embedding space and from the embedding space to the output space, respectively. For each $p \in \{0.01, 0.1, 0.3, 1\}$ models are trained using ordinary and myopic descent on one epoch of 30M sequences of length 64 sampled from $\mathcal{D}^{64}_{p,10,10}$. See Table 3 for architecture details.

Configuration Key	Value
num_layers	2
num_heads	2
embd_dim	128
n_inner	512
input_dim	2
output_dim	1
activation	relu
attn_pdrop	0
embd_pdrop	0
resid_pdrop	0
lr	1e-3
optimizer	AdamW
weight_decay	0.01
betas	(0.9, 0.999)
scheduler	cosine
warmup	0.01
batch_size	512
seq_length	64
loss_fn	HuberLoss

Table 3: Transformer configuration used when training on synthetic data distribution $\mathcal{D}_p$

D.1.1 Probe details

Given a transformer model trained on $\mathcal{D}_p$, we sample the hidden states at each layer when the model is given as input 50 000 evaluation sequences from the same distribution $\mathcal{D}_p$. For each layer and targets $\sin(b\mathrm{x}_{n-i})$ for varying $i > 0$, we fit a linear regression model on the hidden state of that layer to the target. The in-sample R^2 of each linear model is then reported in Figure 2. Figure 6 is a visualization of the linear probe's performance on the vanilla transformer.

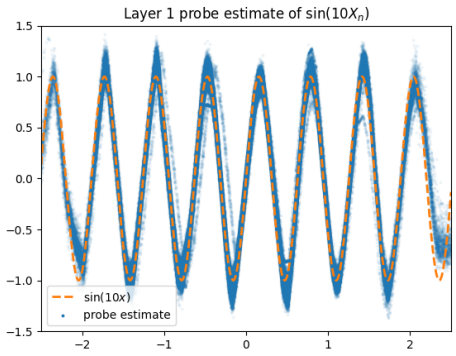


Figure 6: Estimate of $\sin(bx_n)$ by linear probe fit on layer 1 of transformer with vanilla training on $\mathcal{D}_{0.3}$. Computed over 50 000 samples from $\mathcal{D}_1$.

D.2 Natural language setting

D.2.1 GPT-2

For both vanilla and myopic training, we train the GPT2-small architecture from random initialization for one epoch on 4.6M sequences from the MS MARCO dataset (Nguyen et al., 2016), truncated to length 64. To estimate the local myopia bonus of the vanilla model, we train another model from random initialization with the same architecture, but with past hidden states provided by the vanilla model. Note that this "local myopic" model attains slightly *better* performance than the vanilla model; each forward pass can focus purely on next-token prediction, since past hidden states are supplied by a separate model. As a baseline, we also train a "transformer bigram" model, which is a transformer model whose key/value states are zeroed out during training and evaluation. See Table 4 for configuration details.

Configuration Key	Value
num_layers	12
num_heads	12
embd_dim	768
n_inner	3072
vocab_size	50257
activation	gelu_new
attn_pdrop	0.1
embd_pdrop	0.1
resid_pdrop	0.1
lr	6.0×10^{-4}
optimizer	AdamW
weight_decay	0.01
betas	(0.9, 0.999)
scheduler	cosine
warmup	0.01
batch_size	512
seq_length	64
loss_fn	CrossEntropy

Table 4: GPT-2 small configuration used when training on natural language data.